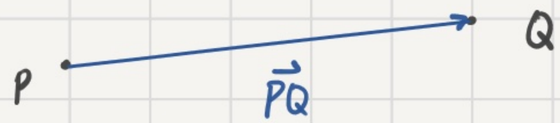
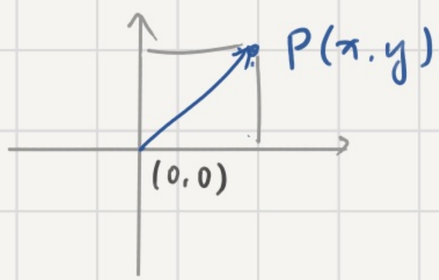
Vectors

Two points P, Q define a vector $\vec{PQ}$:

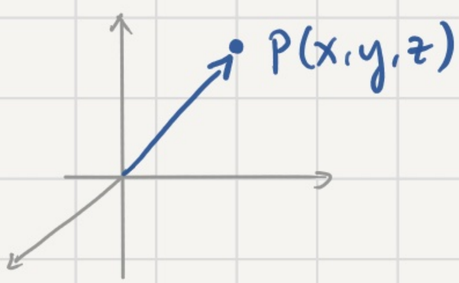


On the xy-plane $\mathbb{R}^2$, a point P can be represented as a vector:



$$\vec{OP} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Similarly, in xyz-space $\mathbb{R}^3$,



$$\vec{OP} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

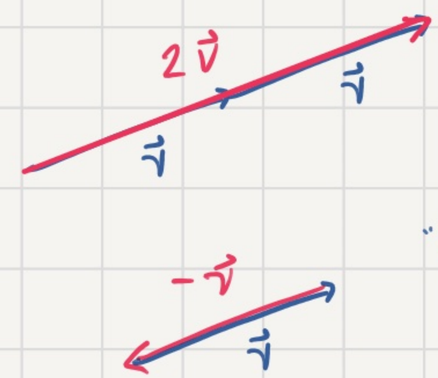
* Vectors indicate quantities with a magnitude and a direction.

Addition of Vectors

$$\vec{v} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \vec{w} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$$\vec{v} + \vec{w} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

Scalar Multiplication



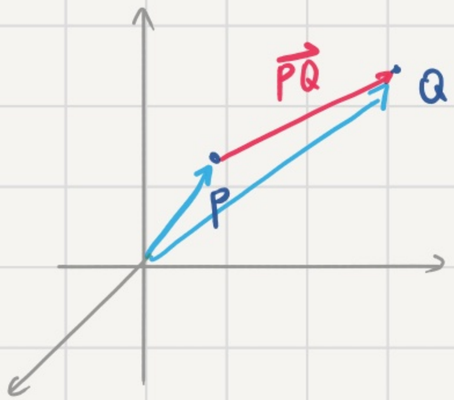
Given a scalar λ,

$$\lambda \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}.$$

"lambda"

Q. What is $\vec{v} - \vec{w}$? $\vec{v} + (-1)\vec{w}$

Q. What is the vector between two points?

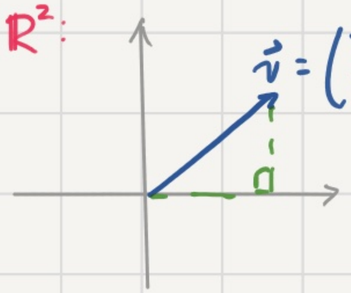


$$\vec{PQ} = \vec{OQ} - \vec{OP}$$

Geometry and Distance

The magnitude (or length) of a vector is

$\mathbb{R}^2$:



$$|\vec{v}| = \sqrt{x^2 + y^2}.$$

$\mathbb{R}^3$: $|\vec{v}| = \sqrt{x^2 + y^2 + z^2}$ for $\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

A vector of length 1 is called a unit vector.

Ex. What is the length of

$$\vec{v} = \begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix}?$$

$$|\vec{v}| = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{169} = 13.$$

We can convert $\vec{v}$ to a unit vector in the same direction

$$\frac{1}{|\vec{v}|}\vec{v} = \frac{1}{13}\begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix} = \begin{pmatrix} 3/13 \\ 4/13 \\ 12/13 \end{pmatrix}.$$

The standard basis vectors are

$\mathbb{R}^2$: $\vec{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\vec{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\mathbb{R}^3$: $\vec{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $\vec{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\vec{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Any vector is a combination of the standard basis vectors!

Ex. $\begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix} = 3\vec{i} + 4\vec{j} + 12\vec{k}$

The zero vector is $\mathbb{R}^2$: $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbb{R}^3$: $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

* This has no direction!

Dot Product

The dot product is the number given by

$\mathbb{R}^2$:

$$\vec{v} \cdot \vec{w} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \cdot \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = x_1 x_2 + y_1 y_2$$

$\mathbb{R}^3$:

$$\vec{v} \cdot \vec{w} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \cdot \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = x_1 x_2 + y_1 y_2 + z_1 z_2$$

If we compute the dot product $\vec{v} \cdot \vec{v}$:

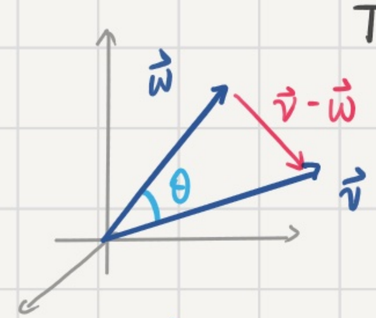
$$\vec{v} \cdot \vec{v} = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = x^2 + y^2 = |\vec{v}|^2$$

$$\boxed{|\vec{v}|^2 = \vec{v} \cdot \vec{v}}$$

Properties. Let $\vec{u}, \vec{v}, \vec{w}$ be vectors. Then:

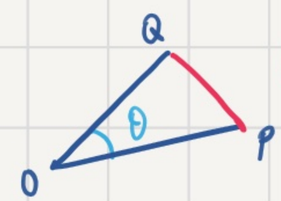
1. $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$
2. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
3. for $\lambda \in \mathbb{R}$: $\lambda(\vec{v} \cdot \vec{w}) = (\lambda \vec{v}) \cdot \vec{w} = \vec{v} \cdot (\lambda \vec{w})$
4. $\vec{0} \cdot \vec{v} = 0$

Theorem. Let θ be the angle between $\vec{v}, \vec{w}$. Then,



$$\boxed{\vec{v} \cdot \vec{w} = |\vec{v}|\,|\vec{w}| \cos\theta}.$$

Proof. Since OPQ is a triangle,



$$|PQ|^2 = |OP|^2 + |OQ|^2 - 2|OP||OQ|\cos\theta.$$

$$[\, c^2 = a^2 + b^2 - 2ab\cos C \,]$$

$$|\vec{v} - \vec{w}|^2 = |\vec{v}|^2 + |\vec{w}|^2 - 2|\vec{v}||\vec{w}|\cos\theta$$

Since $|\vec{v}|^2 = \vec{v} \cdot \vec{v}$:

$$\begin{aligned} |\vec{v} - \vec{w}|^2 &= (\vec{v} - \vec{w}) \cdot (\vec{v} - \vec{w}) \\ &= (\vec{v} - \vec{w}) \cdot \vec{v} - (\vec{v} - \vec{w}) \cdot \vec{w} \\ &= \vec{v} \cdot \vec{v} - \vec{w} \cdot \vec{v} - \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{w} \\ &= |\vec{v}|^2 - 2\vec{v} \cdot \vec{w} + |\vec{w}|^2 \end{aligned}$$

So,

$$\cancel{|\vec{v}|^2} - \cancel{2}\vec{v} \cdot \vec{w} + \cancel{|\vec{w}|^2} = \cancel{|\vec{v}|^2} + \cancel{|\vec{w}|^2} - \cancel{2}|\vec{v}||\vec{w}|\cos\theta$$

$$\Rightarrow \vec{v} \cdot \vec{w} = |\vec{v}||\vec{w}|\cos\theta.$$

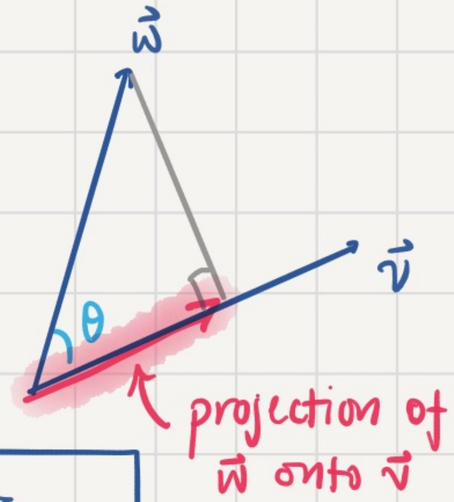
$\square$

Projections

$\star$ $\vec{v}\cdot\vec{w} = |\vec{v}||\vec{w}|\cos\theta$

We have

$$|\vec{w}|\cos\theta = \frac{\vec{v}\cdot\vec{w}}{|\vec{v}|}.$$



The projection vector is

$$\text{proj}_{\vec{v}}\,\vec{w} = \left(\frac{\vec{v}\cdot\vec{w}}{|\vec{v}|}\right)\frac{\vec{v}}{|\vec{v}|}.$$

"the projection of $\vec{w}$ onto $\vec{v}$"

Ex. Find the projection of $\vec{w} = \vec{i} + \vec{j} + 2\vec{k}$ onto $\vec{v} = -2\vec{i} + 3\vec{j} + \vec{k}$.

$$\vec{w} = \begin{pmatrix}1\\1\\2\end{pmatrix} \qquad \vec{v} = \begin{pmatrix}-2\\3\\1\end{pmatrix}$$

$$\vec{v}\cdot\vec{w} = 1\cdot -2 + 1\cdot 3 + 2\cdot 1 = 3$$

$$|\vec{v}| = \sqrt{(-2)^2 + 3^2 + 1^2} = \sqrt{14}.$$

$$\Rightarrow \text{proj}_{\vec{v}}\vec{w} = \left(\frac{3}{\sqrt{14}}\right)\frac{\vec{v}}{\sqrt{14}} = \frac{3}{14}\begin{pmatrix}-2\\3\\1\end{pmatrix} = \begin{pmatrix}-3/7\\9/14\\3/14\end{pmatrix}.$$

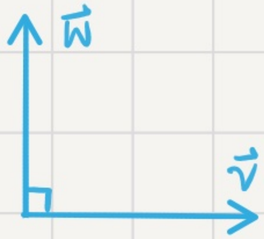
Dot Product:

$$\vec{v}\cdot\vec{w} = \begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}\cdot\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix} = x_1x_2 + y_1y_2 + z_1z_2$$

$$\vec{v}\cdot\vec{v} = |\vec{v}|^2$$

$$\vec{v}\cdot\vec{w} = |\vec{v}||\vec{w}|\cos\theta.$$

If $\vec{v}, \vec{w}$ are perpendicular:



$$\cos\left(\frac{\pi}{2}\right) = 0 \Rightarrow \vec{v}\cdot\vec{w} = 0$$

Two vectors $\vec{v}, \vec{w}$ are orthogonal if

$$\vec{v}\cdot\vec{w} = 0.$$

All vectors are orthogonal to the zero vector.

Cross Product [only in $\mathbb{R}^3$!]

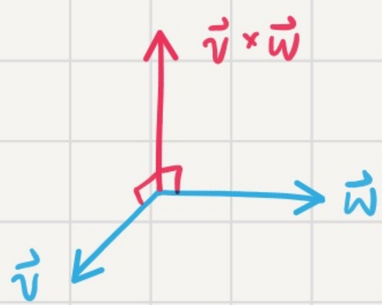
Given two vectors $\vec{v}, \vec{w}$, the cross product is the vector given by

$$\vec{v} \times \vec{w} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \times \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} y_1 z_2 - z_1 y_2 \\ z_1 x_2 - x_1 z_2 \\ x_1 y_2 - y_1 x_2 \end{pmatrix}.$$

Ex. Compute

$$\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ 7 \\ -5 \end{pmatrix} = \begin{pmatrix} 3(-5) - 7(-4) \\ 4(2) - 1(-5) \\ 1(7) - 3(2) \end{pmatrix} = \begin{pmatrix} -43 \\ 13 \\ 1 \end{pmatrix}.$$

Theorem. The vector $\vec{v} \times \vec{w}$ is orthogonal to both vectors $\vec{v}$ and $\vec{w}$.



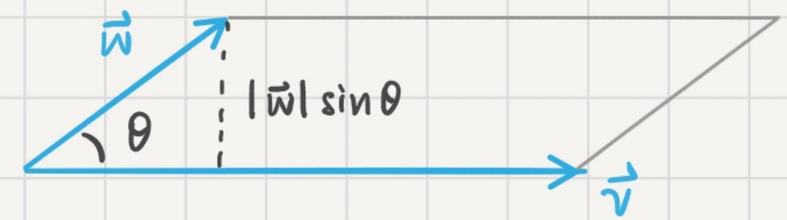
To prove this, show that

$$(\vec{v} \times \vec{w}) \cdot \vec{v} = 0, \quad (\vec{v} \times \vec{w}) \cdot \vec{w} = 0.$$

Theorem. If θ is the angle between $\vec{v}, \vec{w}$:

$$|\vec{v} \times \vec{w}| = |\vec{v}||\vec{w}| \sin\theta.$$

To prove this: use $\vec{v} \cdot \vec{v} = |\vec{v}|^2$.



Thus, $|\vec{v} \times \vec{w}|$ gives us the area of the parallelogram formed by $\vec{v}$ and $\vec{w}$.

Properties. Let $\vec{u}, \vec{v}, \vec{w}$ be vectors in $\mathbb{R}^3$.

1. $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$
2. for $\lambda \in \mathbb{R}$: $\lambda(\vec{v} \times \vec{w}) = (\lambda \vec{v}) \times \vec{w} = \vec{v} \times (\lambda \vec{w})$
3. $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$
4. $(\vec{v} + \vec{w}) \times \vec{u} = \vec{v} \times \vec{u} + \vec{w} \times \vec{u}$
5. $\vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w}$
6. $\vec{u} \times (\vec{v} \times \vec{w}) = (\vec{u} \cdot \vec{w}) \vec{v} - (\vec{u} \cdot \vec{v}) \vec{w}$